



bGeigieZen Kit Assembly Manual for V4.x boards

Introduction

The kit is an excellent tool for DIY learning. Assembly can take as little as 15–30 minutes, depending on your level of skill and experience with electronics.

The current board version V4.x is for the M5Stack controllers CoreS3 and Core2 v1.1 loaded with firmware 3.x.x. V4.x boards can be recognized by the absence of an external USB charging module and on/off switch on the top side of the board that was present on earlier versions.

As with any project, please take the time to prepare by gathering the necessary tools, preparing your work space, reading the manual, and turning off distractions before you start.

WARNINGS

- Wear safety glasses when cutting off leads.
- Be gentle with the kit:
 - Some components are fragile.
 - Some components can be damaged by static electricity. If possible, work on an antistatic mat or wear an antistatic wristband connected to earth ground.
- Be very careful not to short circuit an 18650 rechargeable battery. Should it suddenly get very hot, place it in a closed metal container. The battery can start smoking, emit flames and possibly even explode.
- Handling the LND 7317 Geiger-Müller pancake sensor: The face of the sensor is a very delicate thin mica membrane sealing the partial vacuum in the sensor. The membrane is easily punctured and cannot be repaired, permanently damaging the sensor. The sensor is shipped with a 3D printed plastic clip-on cover. Please keep the cover on at all times – there is no need ever to remove the cover.

Getting help

[[TO BE FILLED IN]]

If you run into problems with the build, or you have questions or suggestions, please post to the Google Groups forum [Safecast Device Discussions and Support](#).



Tools and workspace

Tools needed:

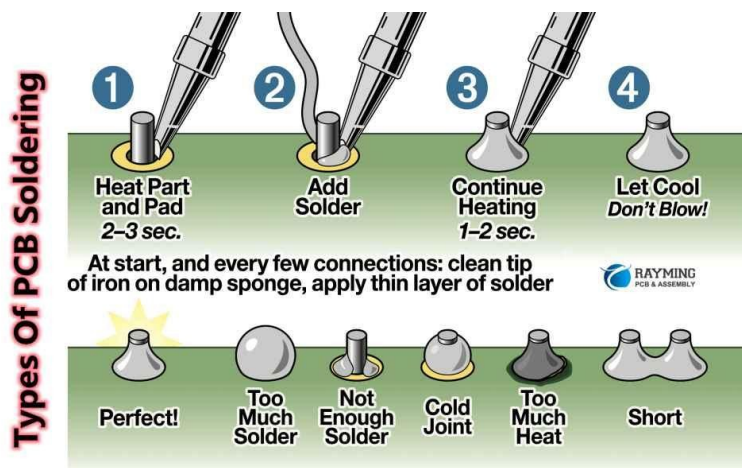
- Flush cutters and nippers (for cutting pin headers and excess wire)
- Temperature controlled soldering iron (fine tip)
- Small Phillips-head Screwdriver
- Small flat-head screwdriver
- USB-C cable for programming and charging (not included in the kit)

Workspace:

- Well-lit workbench
- Silicone rubber mat (antistatic)
- Ventilation to dissipate soldering fumes

Soldering tips

- Please avoid tin-lead (Sn-Pb) solder wire. Although tin-lead (63-37 or 60-40) has historically been widely used in the electronics industry, lead is hazardous to health and is a toxic environmental pollutant. The recommendation is to use lead-free (a.k.a. “No Pb”) soldering wire with “no-clean” rosin, for example MG Chemicals 4900, AIM LV05, Kester 275, etc. made from SAC305 (Sn-Ag-Cu) industry standard alloy. With “no-clean” flux, no cleaning is required.
- If using Sn-Pb solder with rosin flux, use isopropyl alcohol (IPA) to clean the solder joints after soldering.
- Fumes from soldering are harmful for your health, so work in a well-ventilated area. Washing your hands after soldering is advised. Please do not eat/drink while soldering.
- The temperature of the soldering iron should be 371°C to 400°C (700°F to 750°F) for lead-free (No-Pb) alloys. For tin-lead, the melting point is 183°C (361°F).



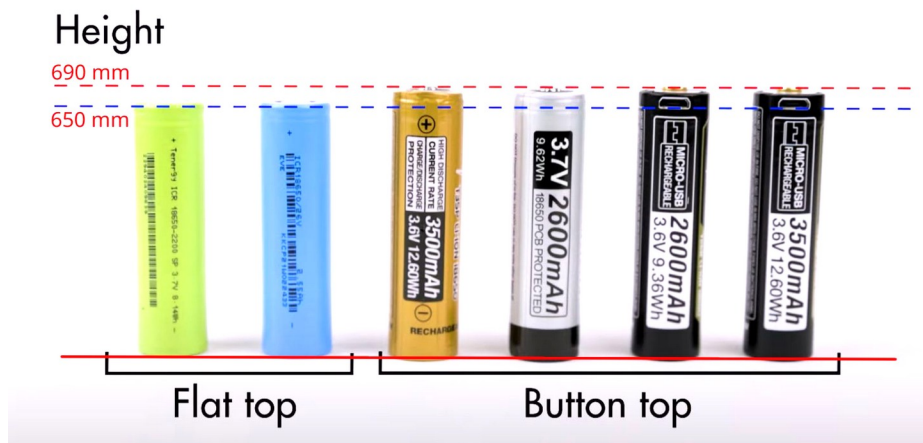


Li-Ion batteries

The bGeigieZen is powered by a 3.7V 18650 type rechargeable lithium-ion (Li-Ion) battery. For shipment safety reasons, the battery is not included in the kit.

The recommended battery is a high capacity Vapecell N40 4,000mAh flat top, though any 18650 type with a flat top can be substituted, e.g. Panasonic NCR18650B or models from LG and Samsung. These batteries are available from online sellers, electronics suppliers and “vape” e-cigarette shops. The battery holder soldered onto the Zen circuit board is meant for a 65 mm unprotected flat top type. If the battery holder on your Zen circuit board has 69 mm spacing, use a protected battery with a button top.

The circuit board incorporates a reverse battery protection diode and a safety fuse to prevent fires due to reverse insertion of the battery.





Parts Inventory

Before you begin assembling the kit, verify that you have all the parts. Contact Safecast if any are missing.

The kit consists of the Pelican 1015 case containing the controller module, sensor and printed circuit board (PCB), and a bag of small parts.

LND 7317 Geiger-Müller pancake sensor

The sensor is charged to a high voltage and triggers a pulse when a radiation event causes an electrical discharge in the tube. The large terminal (post) is the anode, and the long wire is the cathode.

Handling Precautions: The face of the sensor is a very delicate thin mica membrane sealing the partial vacuum in the sensor. The membrane is easily punctured and cannot be repaired, permanently damaging the sensor. The sensor is shipped with a 3D printed plastic clip-on cover. Please keep the cover on at all times – there is no need ever to remove the cover.



Figure 1: LND-7317 Geiger-Müller pancake sensor with protective cover. The post is the anode and the wire is the cathode.



M5Stack CoreS3 Controller

The controller provides the display and user interface through a 2-inch (5.04 cm) touch sensitive colour TFT screen. The controller includes a power management IC for On/Off control, battery percentage indicator and to regulate the battery charging. The connector connects to the PCB through the dual row of pins to receive data from the sensor and the GPS module, and power from the battery.

Note: some older kits use the M5Stack CoreS2 V1.1. The modules are functionally the same and run the same firmware. The only difference is the back cover on the Core2 which must be removed with the small Allen key provided with the kit.

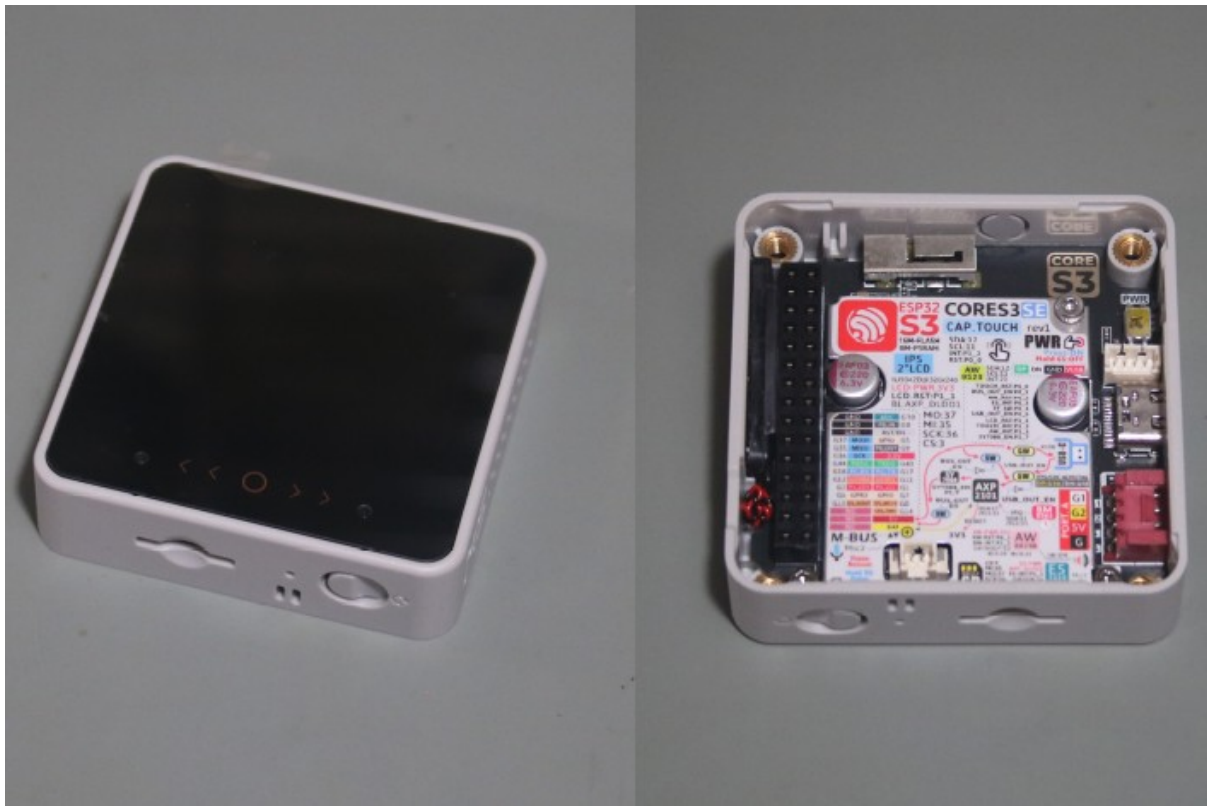


Figure 2: M5Stack CoreSE3 controller. The 2×15 connector is at the left edge.

Printed circuit board (PCB) V4.x

The circuit board carries the system components and connects them through soldered connections.

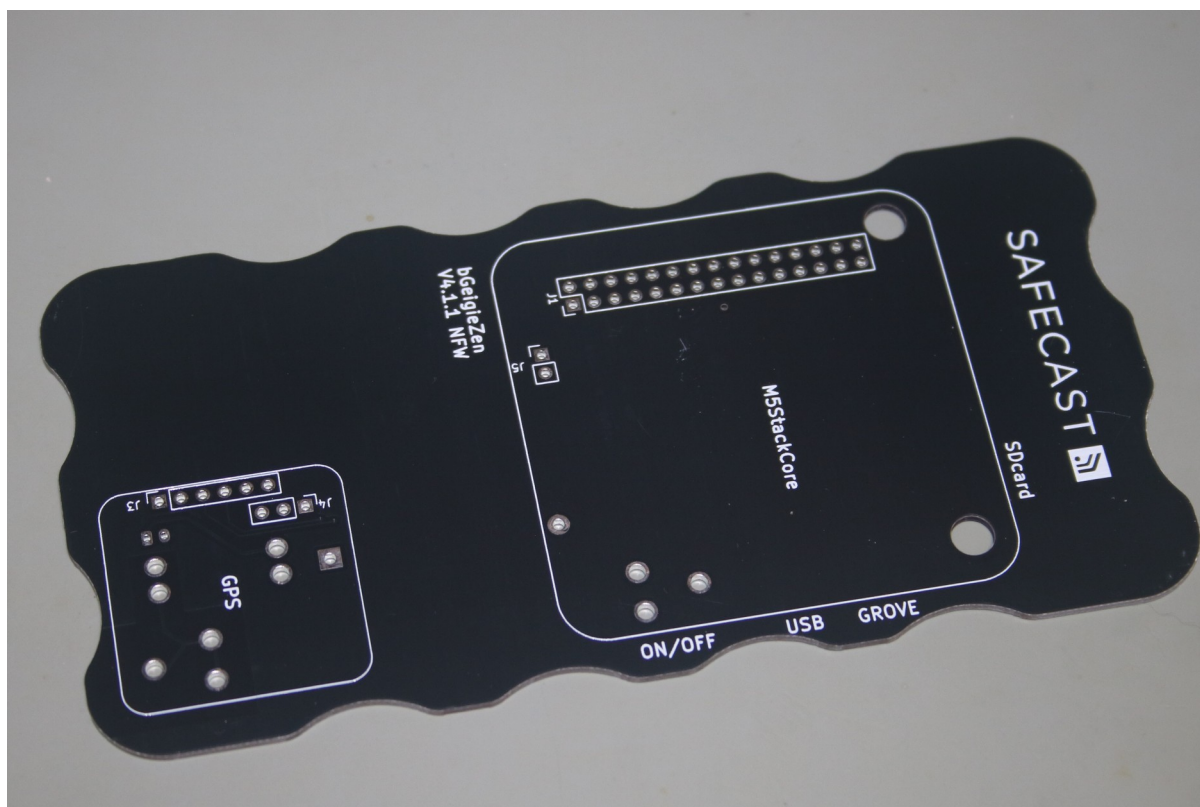


Figure 3: Top side of the printed circuit board.

Pelican 1015 micro case

The transparent polycarbonate case provides protection for the instrument. The thermoplastic rubber liner has been trimmed for the sensor and the battery. Note: the case is watertight but not meant to be submerged.



Figure 4: Pelican 1015 Micro-Case with liner cutouts for the sensor and battery.

Small parts

The small bag contains parts to be assembled on the circuit board. The bag also includes the MicroSD card preconfigured with the operating parameters.

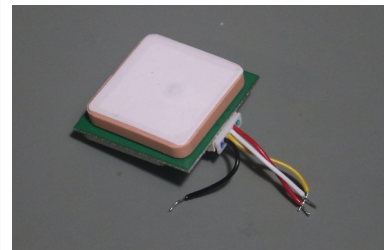
Caution: some parts are static sensitive and can be damaged by static electricity discharges. If available, wear an antistatic strap or work on an antistatic surface.



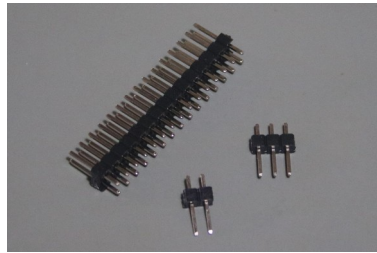
Charger cable



Sensor high voltage wire



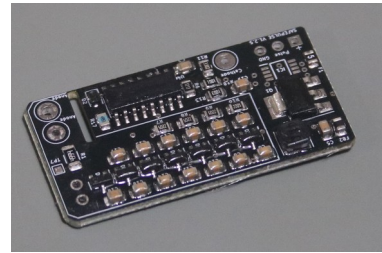
GPS receiver module



Pin headers



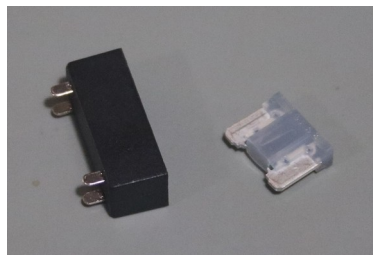
Battery protection diode



Safepulse 500V source



Battery holder clips



Fuse holder and fuse



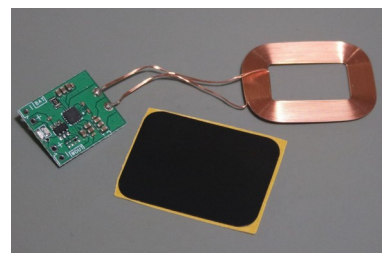
M3 Phillips truss head (x2)



Preconfigured MicroSD card



Heat shrink tubing



Wireless charge receiver

Kit Assembly

Follow the instructions to assemble your bGeigieZen. Observe the photos carefully as you read the instructions.

M5Stack controller header

The controller is assembled onto the printed circuit board (PCB) with the 2 row by 15 position header in multiple stages.

1. Dry fit the M5 CoreS3 controller module to the PCB
 - a. Place the 2×15 header into the holes on the PCB, long side up.
 - b. Carefully line up the header pins with the corresponding sockets in the CoreS3 board
 - c. When the pins are inserted, press the controller module to seat it flush against the top surface of the PCB. Be very careful to correctly line up the pins so that they do not get bent.
 - d. Turn the assembly over to expose the holes for the screws to fasten the controller to the board.
 - e. Insert and *loose*ly tighten the two fastening screws (you will remove the



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- controller later).
2. Solder the pins to the printed circuit board.
 3. Unscrew the controller screws and remove the controller for the remainder of the assembly steps. Some of the solder joints will be hidden under the controller in the final assembly.

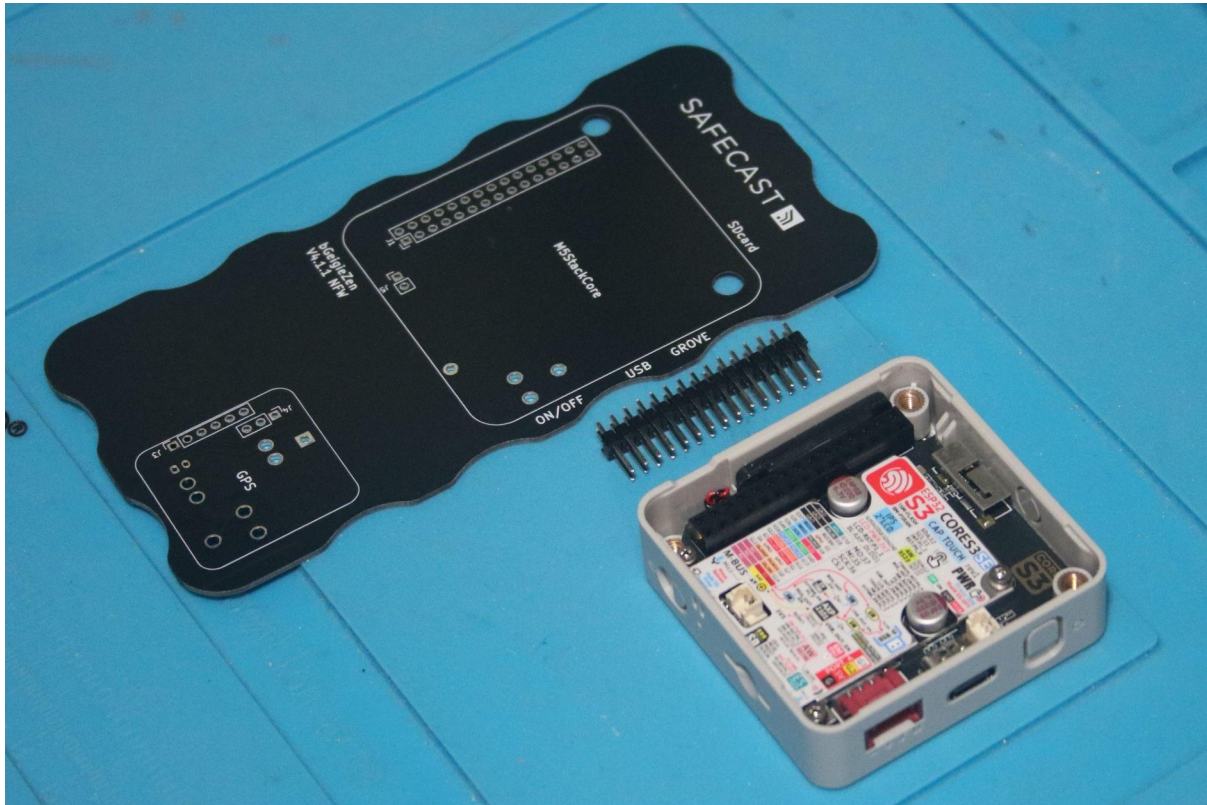


Figure 5: M5Stack CoreS3, 2×15 header and bGeigieZen PCB.



Figure 6: Placing and dry fitting the M5Stack CoreS3 controller onto the PCB.

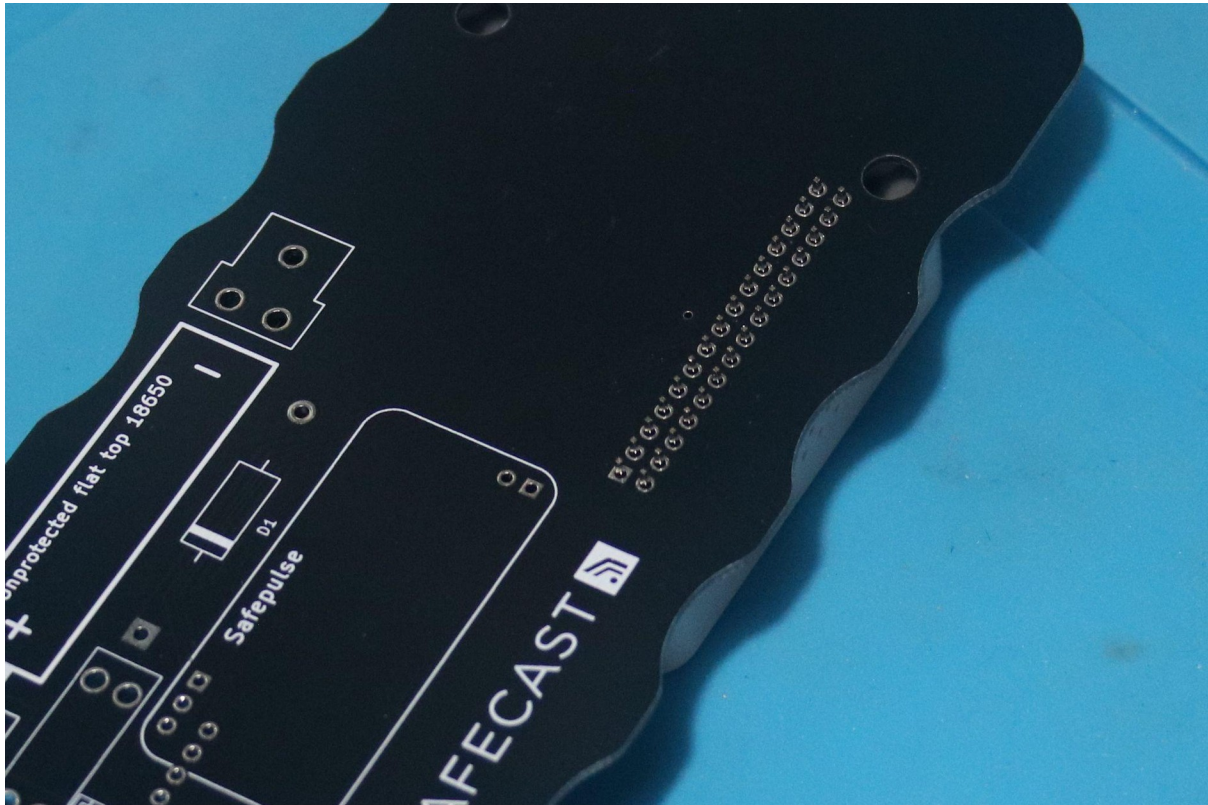


Figure 7: The short end of header pins must protrude from the bottom of the board.

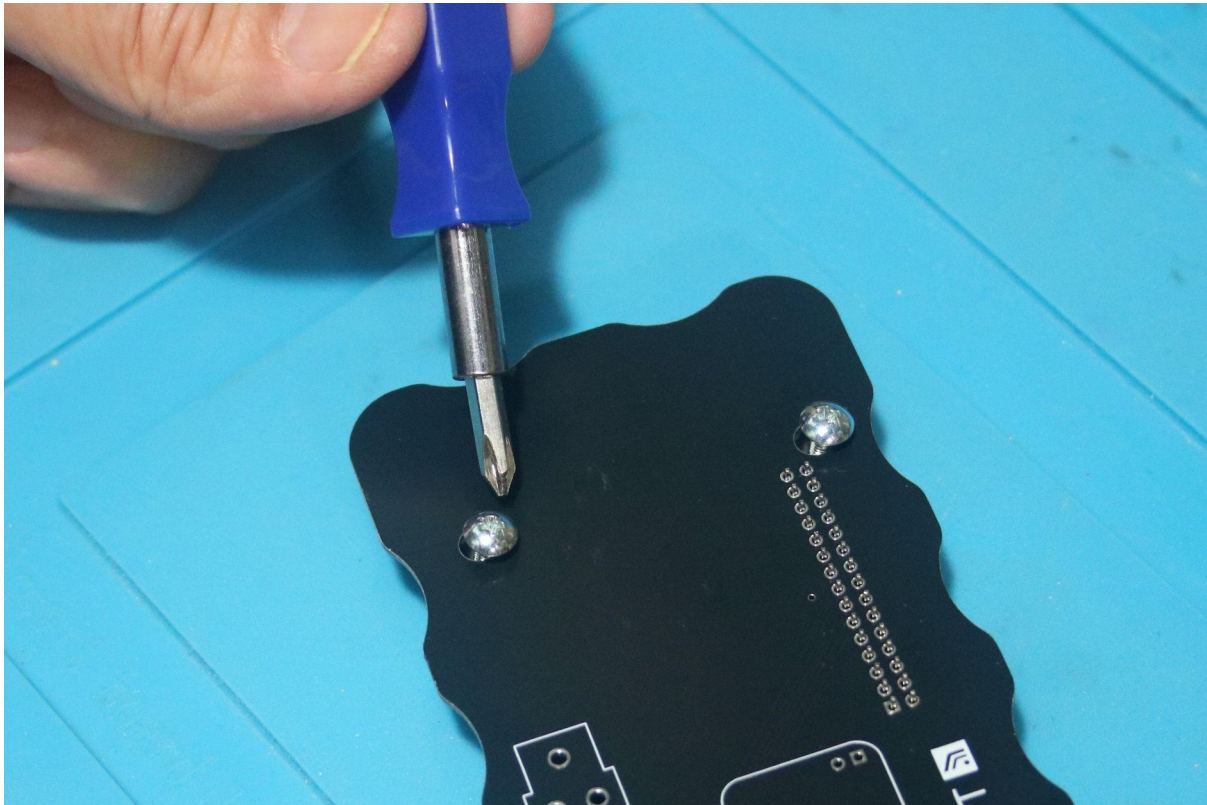


Figure 8: Insert the two screws into the captive nuts in the controller and tighten loosely.

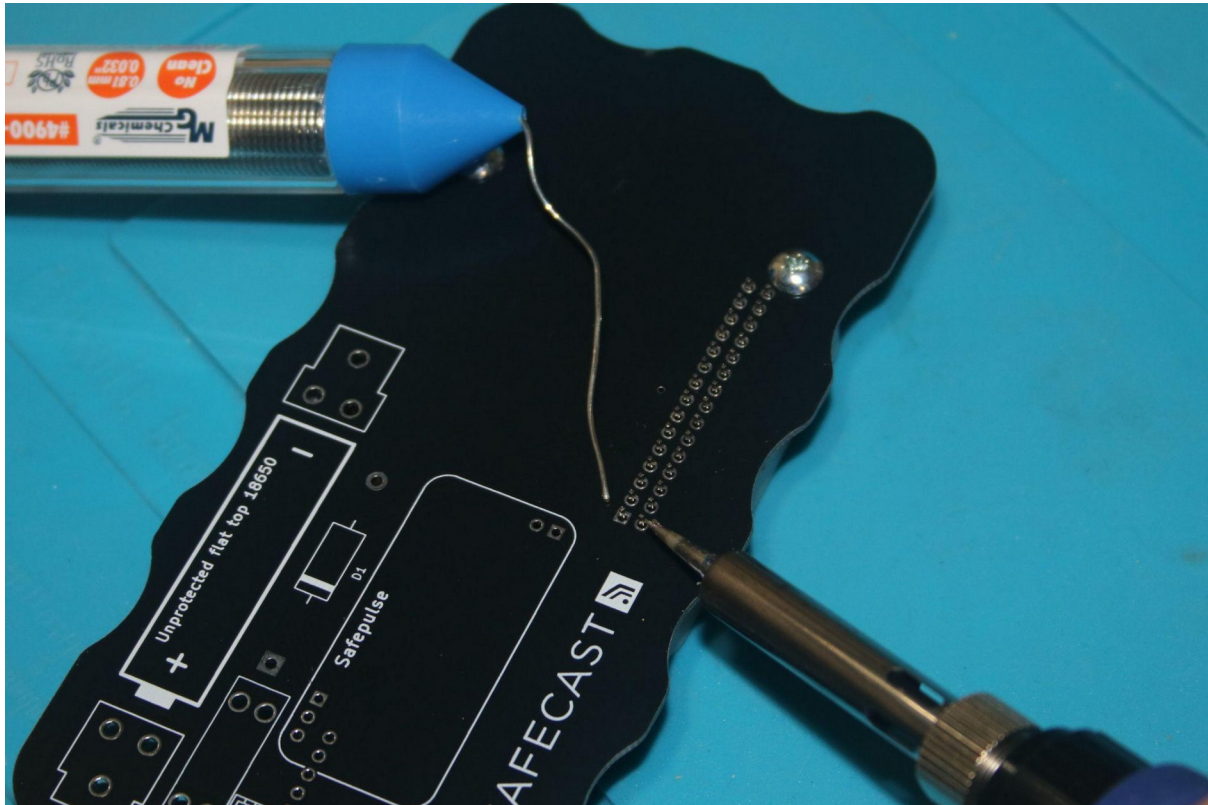


Figure 9: Solder the pins carefully.



Figure 10: The solder joints look like the solder flowed up the pins and around the PCB rings (pads).

Battery clips

The battery clips are assembled onto the bottom of the printed circuit board (PCB) and soldered into place. Make sure that the clips are flush with the surface of the board.

1. Dry fit the battery clips to the PCB
 - A. The clip is assembled on the bottom of the circuit board to match the outline of the 18650 battery.
 - B. The pins of the battery clip should be clenched slightly, a bit like a staple, so that it stays in place when the board is held upside down for soldering.
 - C. The pins should protrude evenly on the top side of the board.
 - D. Make sure that the clip lies flat on the board, and that it does not slip out while soldering.
2. The battery clip pins require a lot of heat to get hot enough for the solder to flow. If the solder does not melt and flow readily, you may need a more powerful soldering iron, or apply the tip of a second iron set to the same temperature. The solder must flow evenly over the pins and into the holes. A minor gap in solder coverage is acceptable, as long as there is a continuous fillet for the rest of the pin.



3. Solder the second clip to complete the battery holder assembly.

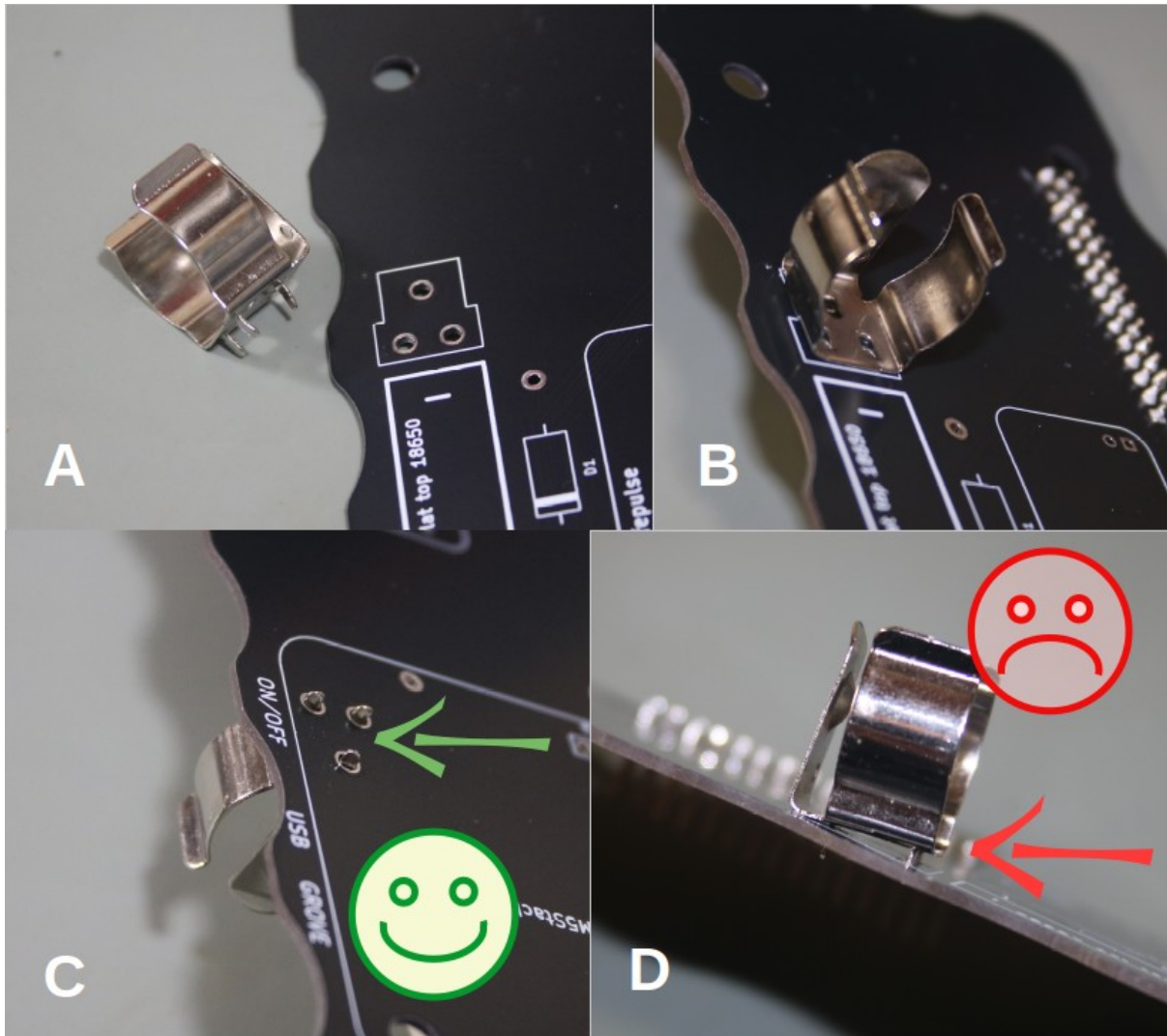


Figure 11: Placing and soldering the battery terminals flush to the board.

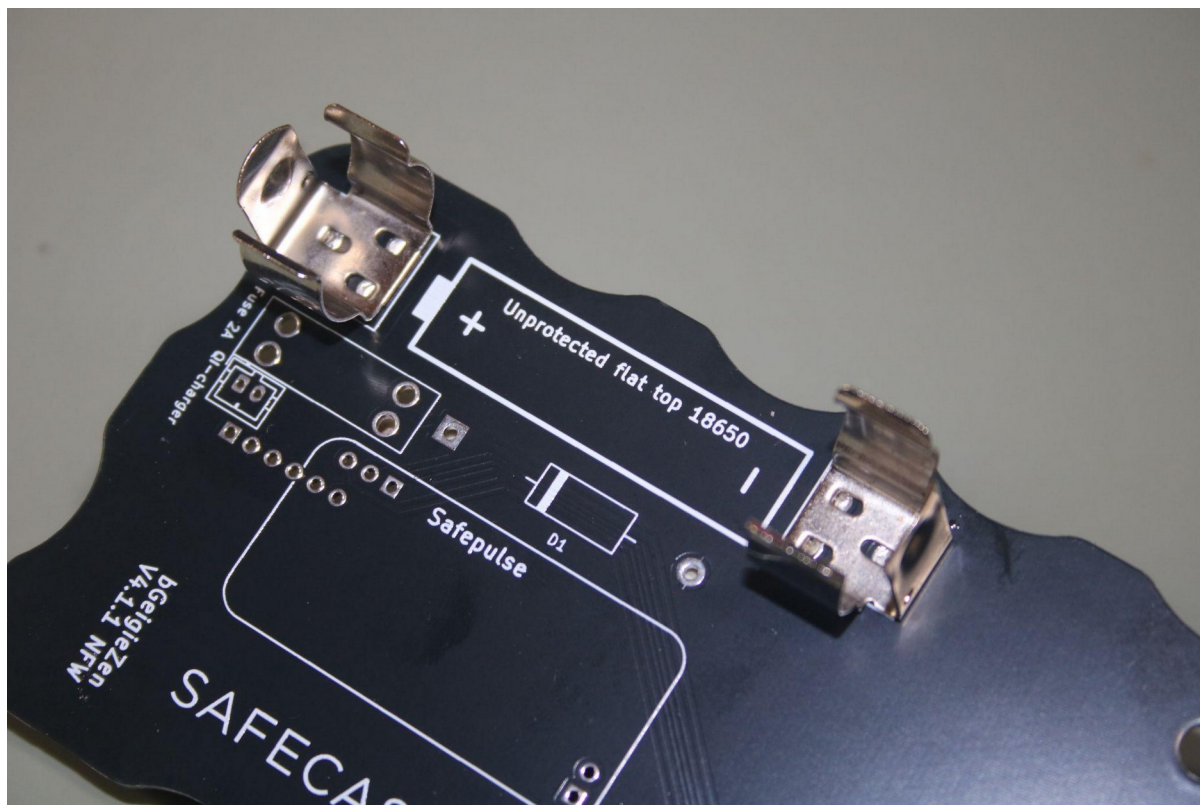


Figure 12: Circuit board with battery terminals.

Protection diode

The rectifier diode, in combination with the fuse, is meant to prevent damage if the battery is inserted backwards. The diode will start conducting fault current, causing the protection fuse to “blow” and halt the current before the battery overheats.

The orientation of the diode must match the outline on the PCB. The white or silver band on the diode body matches the thick line on the outline.

Caution:

1. When bending the leads of the diode, do not put strain on the body itself. Use two pairs of pliers to bend the wire away from the diode body.
 2. Protect your eyes when clipping the excess leads — they can go flying!
- A. Observe the correct orientation of the diode. The white or silver stripe lines up with the thick white line on the board, closest to the hole with the square pad.
 - B. Bend the diode leads to match the holes in the PCB at either end of the outline.
 - C. Insert the leads into the holes and solder from the top of the PCB, the side opposite the diode body.
 - D. Trim the excess leads with a set of flush cutters (a.k.a. side cutters), taking care to prevent the cut ends from flying.

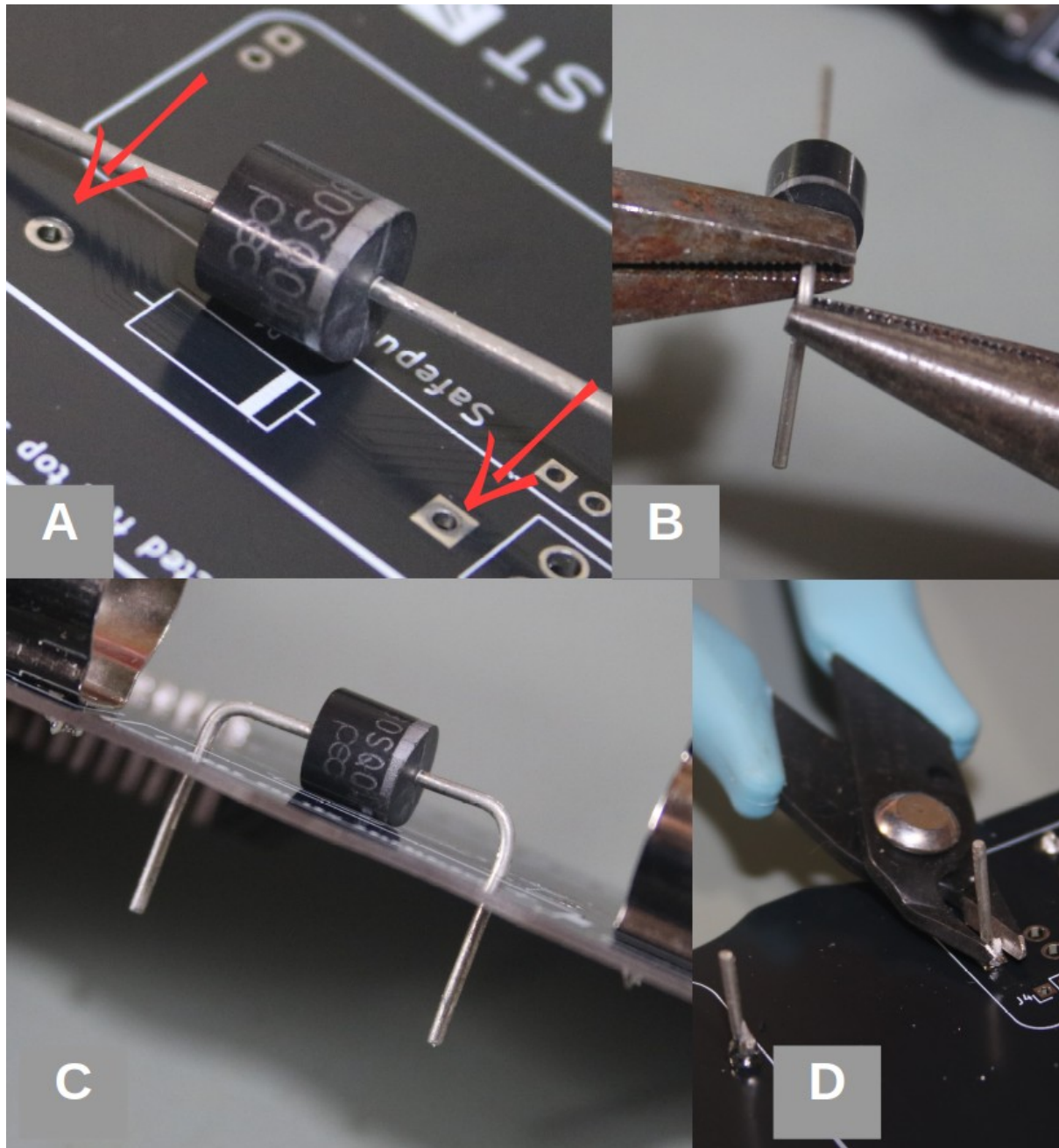


Figure 13: Reverse battery protection diode. A: Orient the diode to match the footprint on the board. B: Bend the leads away from the diode body. C: Diode ready to solder. D: Trim the excess lead wire.



Figure 14: Battery clips soldered in place.

Fuse holder

The fuse, in combination with the rectifier diode, is meant to prevent damage if the battery is inserted backwards. The diode will start conducting fault current, causing the protection fuse to “blow” and halt the current before the battery overheats.

The fuse holder does not have a specific orientation, and the leads are short enough that they do not need to be trimmed after soldering.

Insert the fuse holder into the four holes in the outline marked “FUSE 2A”. The base of the holder must be flush with the surface of the board. The battery clip adjacent to the holder may need to be bent inwards to accommodate the holder.

Solder all four pins, ensuring that the solder flows over the pins and the PCB metal pads.



Figure 15: Fuse holder soldered in place. The fuse will be inserted later.

Charger cable

The bGeigieZen is charged by a non-contact Qi charger – the little board with the wire coil. The charger cable brings power to the board. The cable consists of a receptacle and a plug, already assembled. Insert the pins of the receptacle into the holes in the outline “Qi-charger” such that the red wire is closest to the edge of the board. Use the wires as a handle while soldering the pins to ensure that the receptacle sits flush on the board.

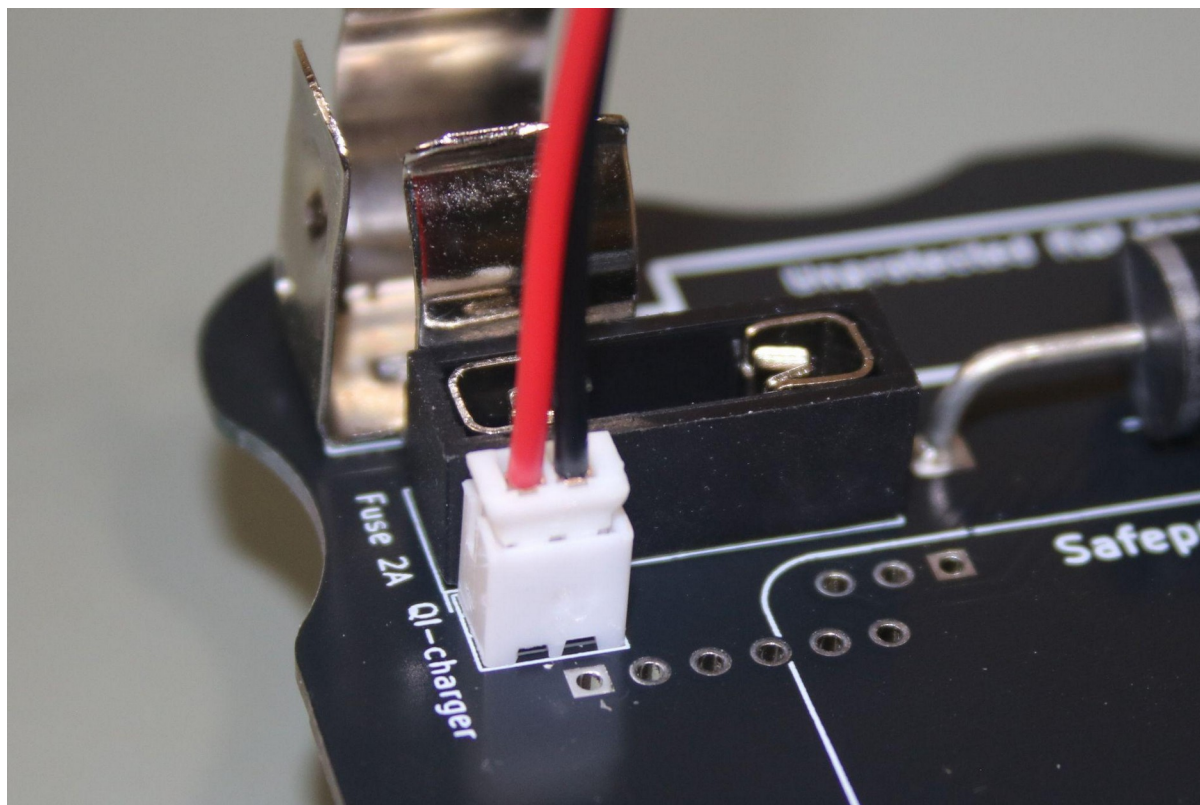


Figure 16: Two-part connector for the battery charger. Observe the correct polarity.

Battery terminal insulation

This step is optional: to prevent accidental battery discharge, potentially causing a fire, the battery terminal blades should be insulated in case the assembled Zen is placed bottom side down on a metal surface. The thickness of the tubing causes the battery to sit high in the clips and push the circuit board up in the case.

The kit includes pre-cut pieces of heat shrinkable tubing. The red pieces are for the positive (+) clip (red is hot!) and the black pieces for the negative (-). Some kits only have black tubing.

Slip the tubing on the sides of the clips (not the end contacts, obviously), leaving a short length 1-2mm above the top of the metal. Apply heat to shrink the tubing over the blades.

Note: If the kit did not come with tubing, obtain a short length (10 cm or so) of 6mm ($\frac{1}{4}$ inch) diameter tubing and cut lengths of 18 to 20 mm ($\frac{3}{4}$ inch).



Figure 17: Cover the battery clip blades with heat-shrinkable tubing to insulate against accidental short circuits when resting on a metallic surface.

Safepulse and GPS receiver

The Safepulse module provides the 500V high voltage supply to the Geiger-Müller sensor and delivers pulses to the controller logic-level input. The GPS receiver provides position and time information for the data logging operations.

CAUTION: Follow the recommended sequence of operations to solder the Safepulse high voltage supply and the GPS receiver modules. The solder connections are on both sides of the circuit board in the same area. Placing either module into place – GPS or Safepulse – before the soldering is complete would prevent installing the other module.

Safepulse pins

The Safepulse is attached to the board through two sets of pin headers: three pins for J4 and two pins for J5 (Figure 18). Place the pins into the J4 and J5 holes such that the long end with plastic carrier is on the bottom of the board (same side as the battery, diode and fuse holder).

GPS receiver wires

The GPS receiver is connected to the circuit board through a set of four wires, then attached using a foam pad with adhesive on both faces (Figure 19). The wires have been prepared to be inserted into the site labelled J3. There are four wires (colours may vary) to be inserted into the four middle holes of J3. Notice that the two outside wires (positions 1 and 6) have been cut short at the module (they are not used in the bGeigieZen). Position the receiver module upside down on the top side of the circuit board, next to J3, then solder the wires. Do not peel off the wax paper for the module adhesive pad. The module will be attached later, after the pins for the Safepulse have been soldered.



Soldering the Safepulse module

Place the Safepulse on the pins, leaving only a short length protruding through the holes, then solder the pins (Figure 20).

Mount the GPS module

Peel the wax paper or plastic tape from the bottom of the GPS module and stick the module in the outline on the circuit board (Figure 21).



Figure 18: Safepulse high voltage supply module mounting pins. The pins face long side up from the bottom side of the board. Do not solder the Safepulse in place until the GPS module wires are soldered.

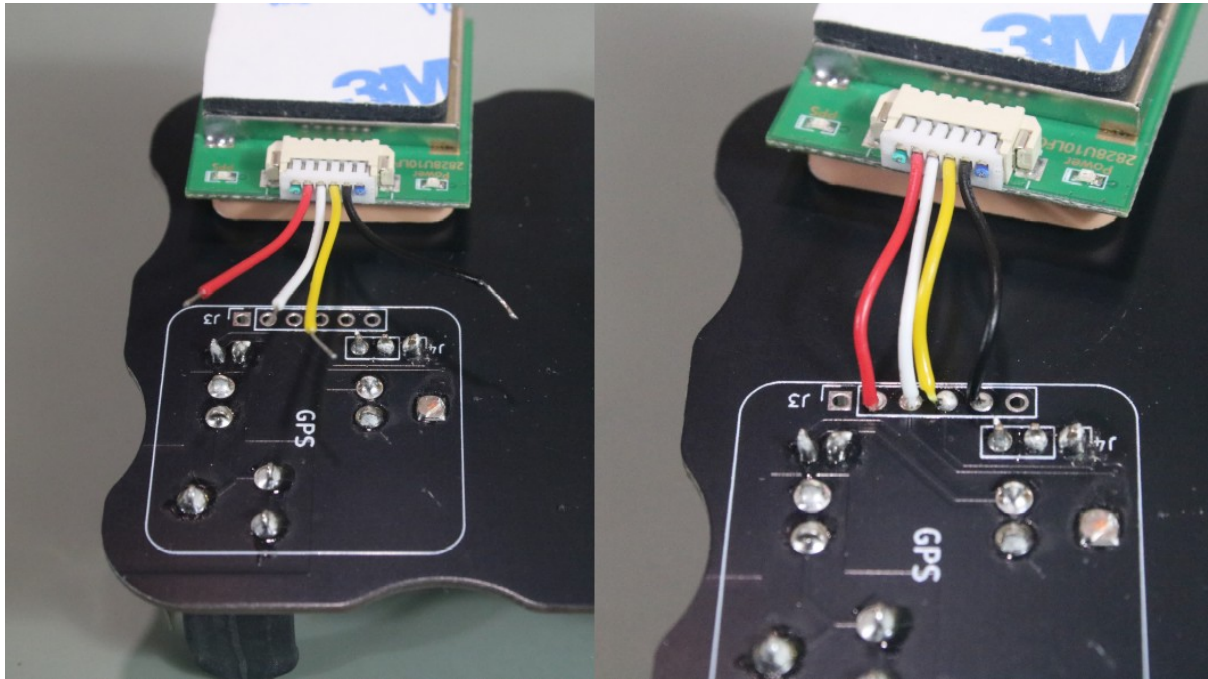


Figure 19: GPS wires soldered into the site for J3. Note that the four wires connect to the four centre holes of J3, leaving the end holes empty.

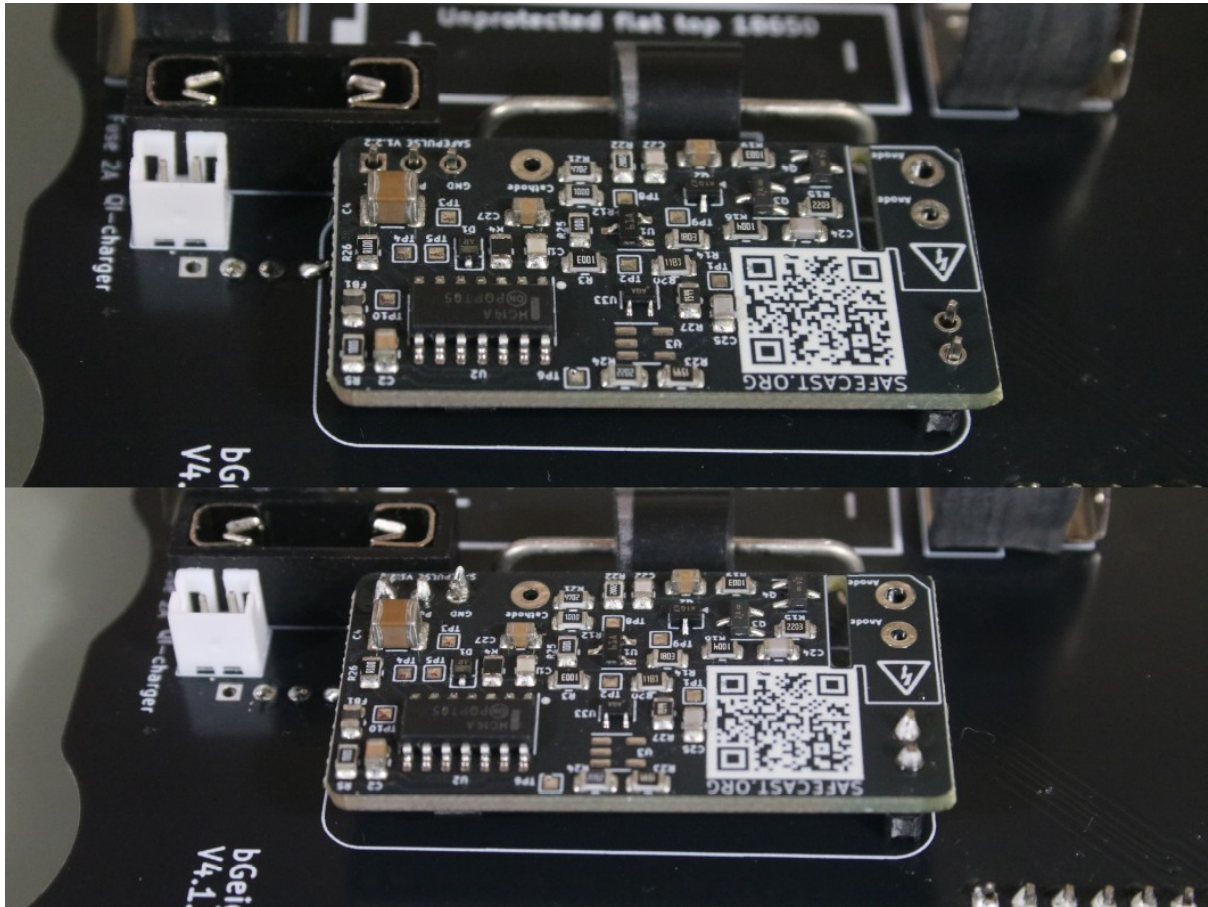


Figure 20: Place the Safepulse module on the mounting pins, keeping the module clear above the main board. Solder in place.

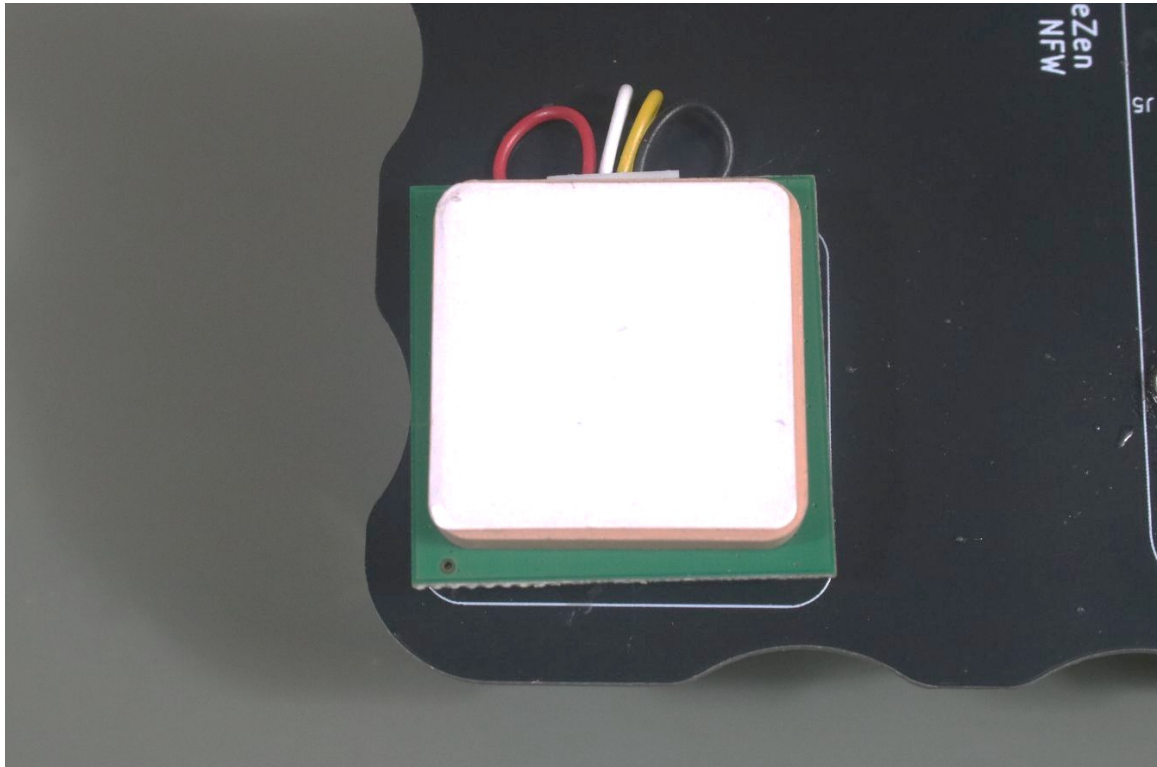


Figure 21: Peel the wax paper from the two-sided mounting pad and press the GPS module onto the board.

M5 CoreS3 Controller

The M5 CoreS3 Controller is mounted on the pin header and fastened with two machine screws. Carefully align the header pins with the socket on the controller's circuit board and press the controller into place.

Secure the controller with the two machine screws. Tighten the screws but DO NOT OVERTORQUE.



Figure 22: Mount the CoreSE3 controller onto the 2×15 pins and secure with the screws.

LND 7317 Geiger-Müller pancake sensor

The Geiger-Müller sensor is placed into position on the back cover of the Pelican 1015 case, then the circuit board is pressed onto the adhesive pad to form a single assembly.

Place the sensor into the back cover of the case (Figure 23). Ensure that the edges of the protective cover are flush against the clear plastic of the cover – there should be no rubber pinched between the sensor cover and the case.



Figure 23: Position the sensor in the bottom liner cutout. Angle the anode post away from where the Safepulse will sit in the final assembly.

The sensor must be angled so that the large anode contact is close to the edge of the case. This ensures enough clearance from the Safepulse HV supply module.

Peel off the wax paper from the adhesive foam pad on the sensor, then place the circuit board into the bottom of the case. When the board is nested into the rubber, press down to attach the sensor to the circuit board (Figure 24).

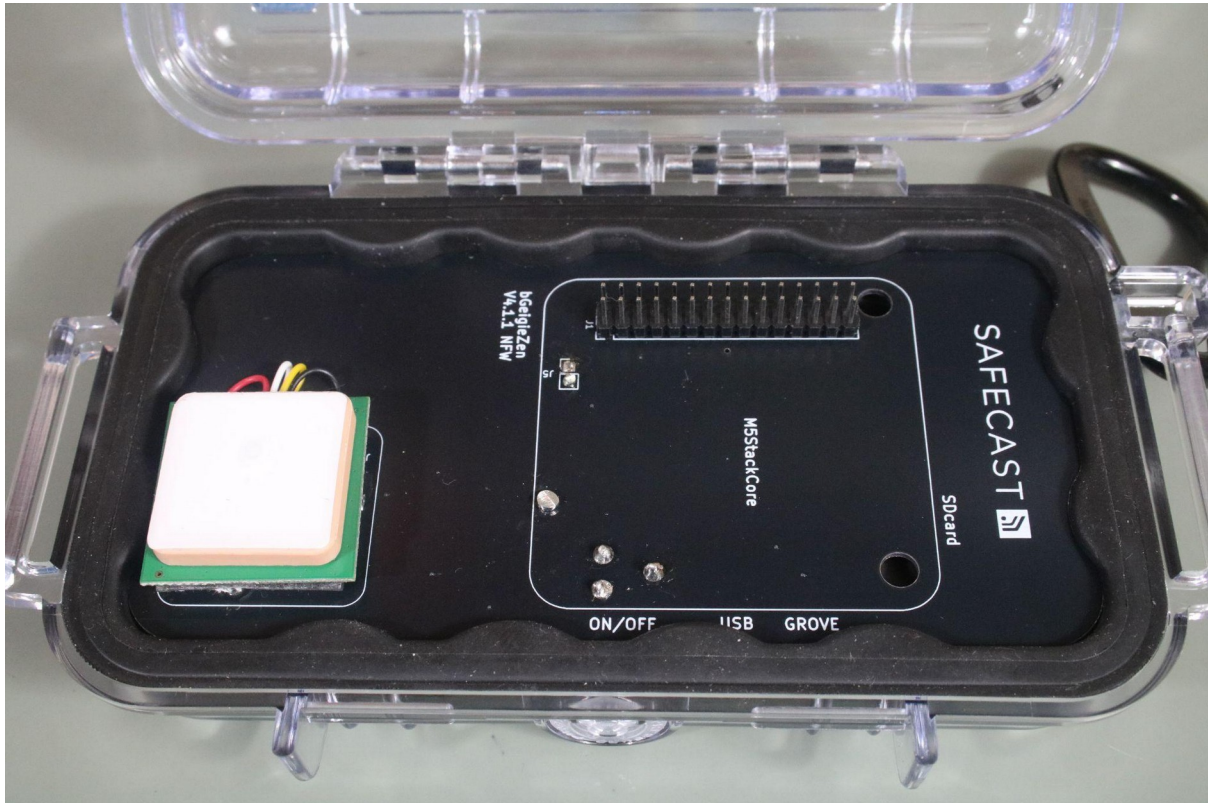


Figure 24: Push the circuit board down to set the adhesive pad.

Remove the entire assembly from the bottom of the case and confirm that the anode post is clear of the Safepulse module (Figure 25).



Figure 25: The sensor is attached to the circuit board. The anode post clears the Safepulse module.

Pancake sensor wiring

The red anode wire connects the sensor anode terminal to the Safepulse (Figure 26). The exposed end of the anode wire must be tinned before soldering to the Safepulse, otherwise some of the wire strands may stay out of the hole and could short out elsewhere on the board. In tinning, the strands of wire are soldered together to make a single solid wire. Plug the red anode socket onto the anode terminal and solder the wire into one of the holes marked "Anode" on the Safepulse. It doesn't matter which of the two, they are connected internally.

Cover the post of the anode with heat shrinkable tubing to prevent accidental contact with the 500V bias voltage.

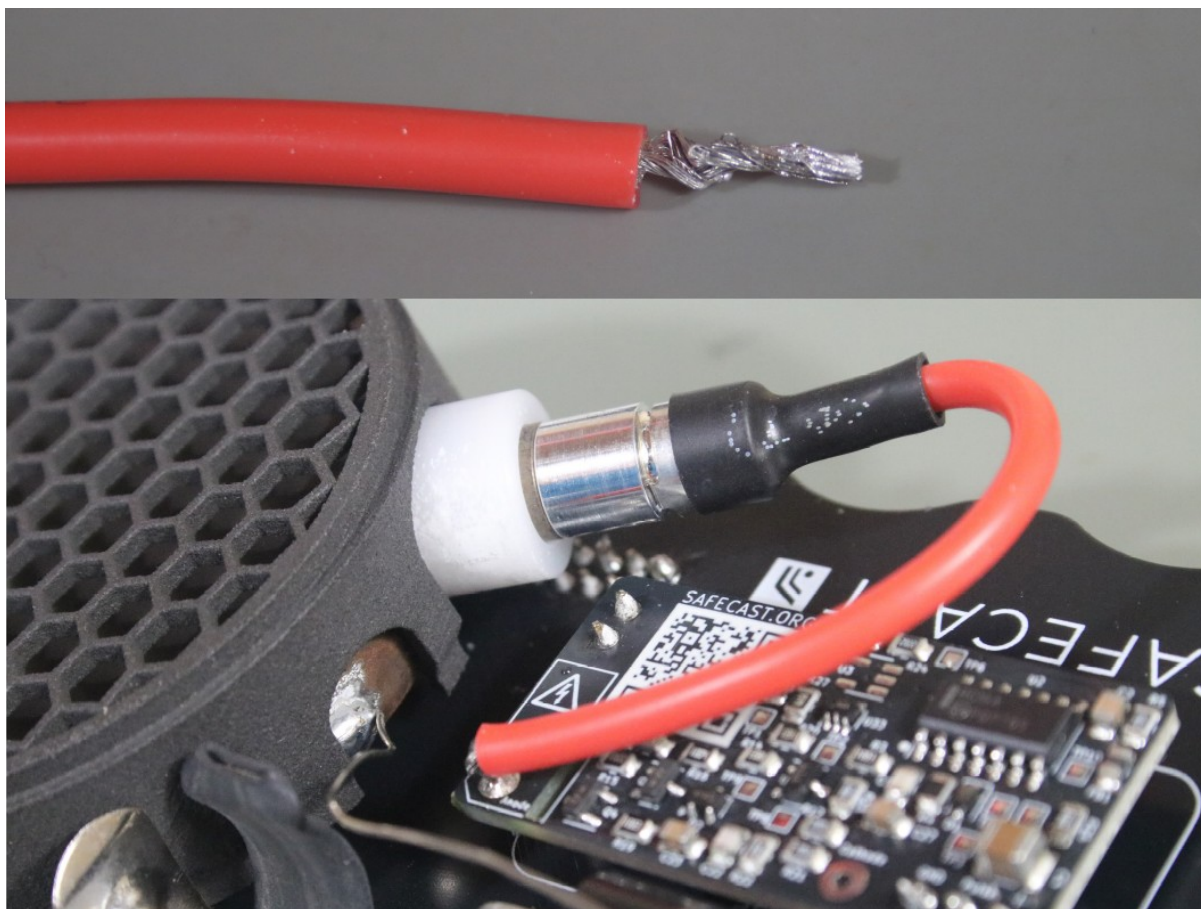


Figure 26: Anode wiring. Top: tin the anode lead before soldering. Bottom: anode lead in board.

Cover the bare cathode wire with the 2 inch (5 cm) length of heat shrinkable tubing. Heat the tubing to shrink it over the wire. Solder the end of the wire into the hole marked “Cathode” (near the pins for J4). You may need to trim any excess length of bare wire before soldering to the Safepulse (Figure 27).

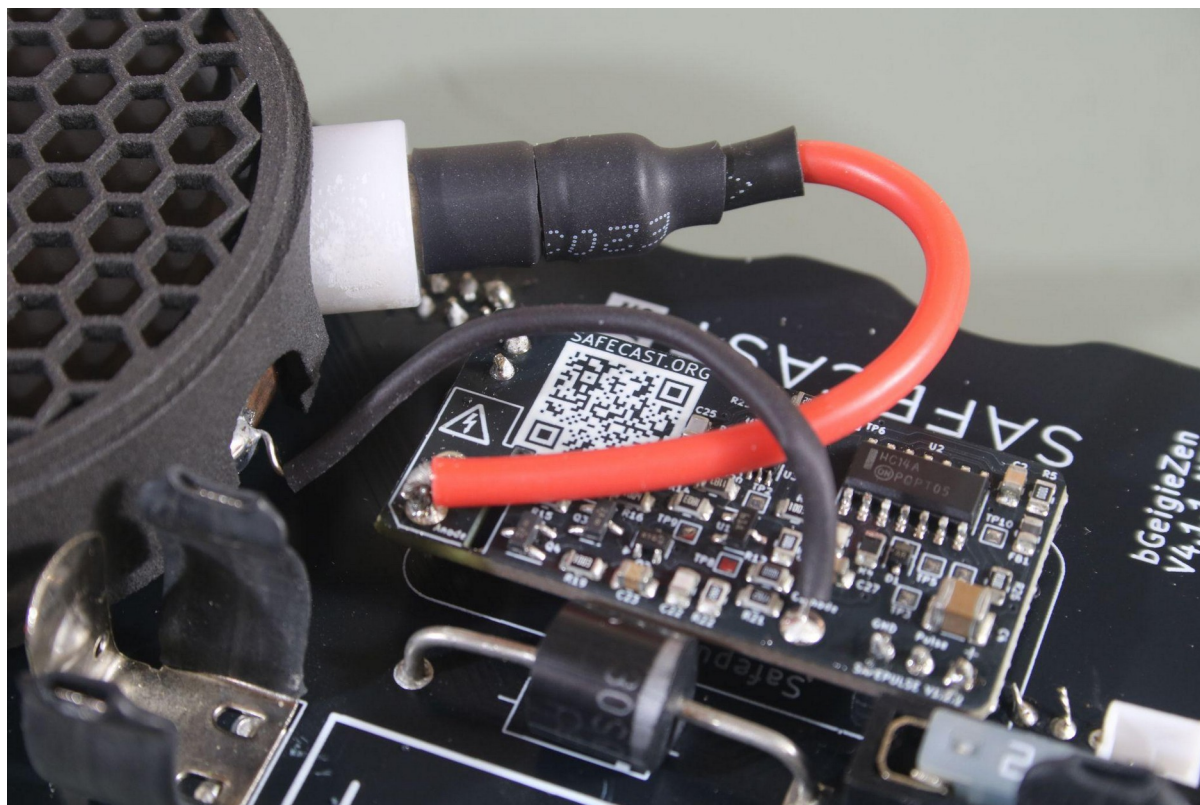


Figure 27: Cathode wire covered with heat shrinkable tubing and soldered into Safepulse. The anode post is also covered with tubing.

Preliminary testing

Before continuing, test the assembly to confirm that the controller and GPS can power up and communicate with each other, and that the controller is receiving pulses from the Geiger-Müller pancake sensor.

Power the controller from a USB power adapter or a USB port on a computer or laptop. The boot-up screen should appear and warn that the SD Card is not present. Ignore the warning and press the Continue button (top right). You should see the Drive screen. Confirm that the sensor is registering counts and that the GPS power light is on, and it is flashing a 1-second LED (it should communicate with the controller even if it does not have a position fix yet). With the minimum functionality verified, continue assembling the device.

Wireless Qi charger

The wireless charger coil and interface module are placed on the underside of the rubber liner, against the bottom of the case (Figure 28). The interface module is placed on the bottom (when facing the complete assembly) side such that the wire from the main board to the BAT + and – terminals goes through the rubber liner.

Punch a hole in the bottom liner to pass the battery wire connector through. Offset the hole



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to keep the module centered (for looks 😊).

Solder the battery connector wires to the interface module BAT + and – holes: red to + (positive) and black to – (negative). Push the connector through the hole (the rubber will stretch). Pull the wire tight so that the interface board is up against the rubber liner (Figure 29).

Peel off the wax paper from the black sticker that came with the coil and apply it to the coil, then apply it face up to the underside of the liner. Stick it to the bottom of the plastic case. The completed bGeigieZen assembly will press the liner against the plastic and hold the coil in place (Figure 30).

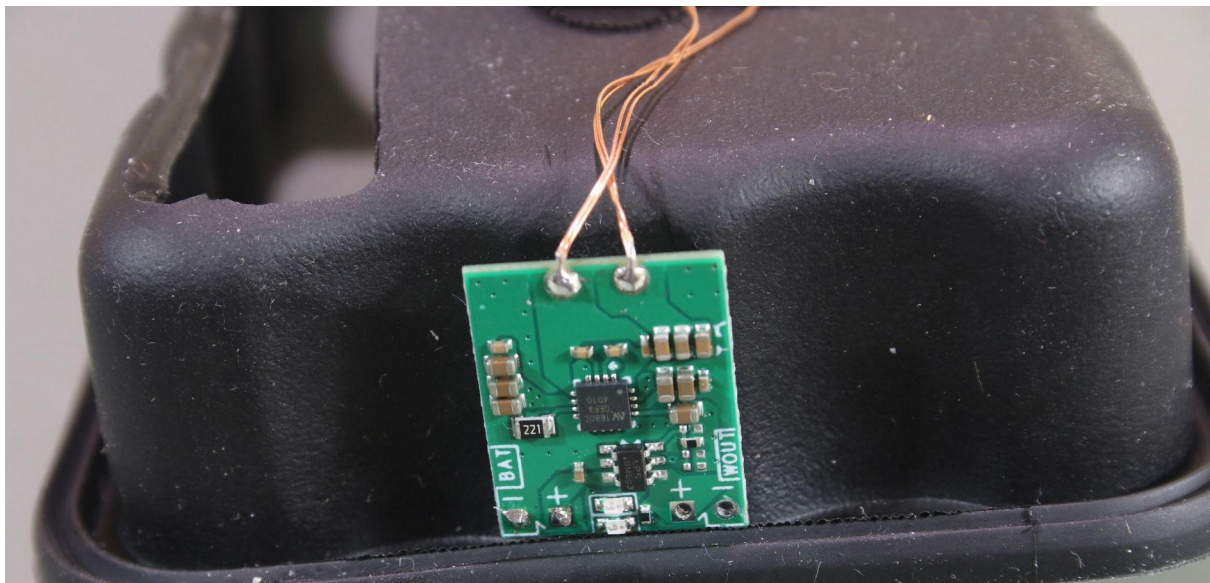


Figure 28: Charger module on the bottom of the case liner. The battery wire is passed through a hole in the rubber.

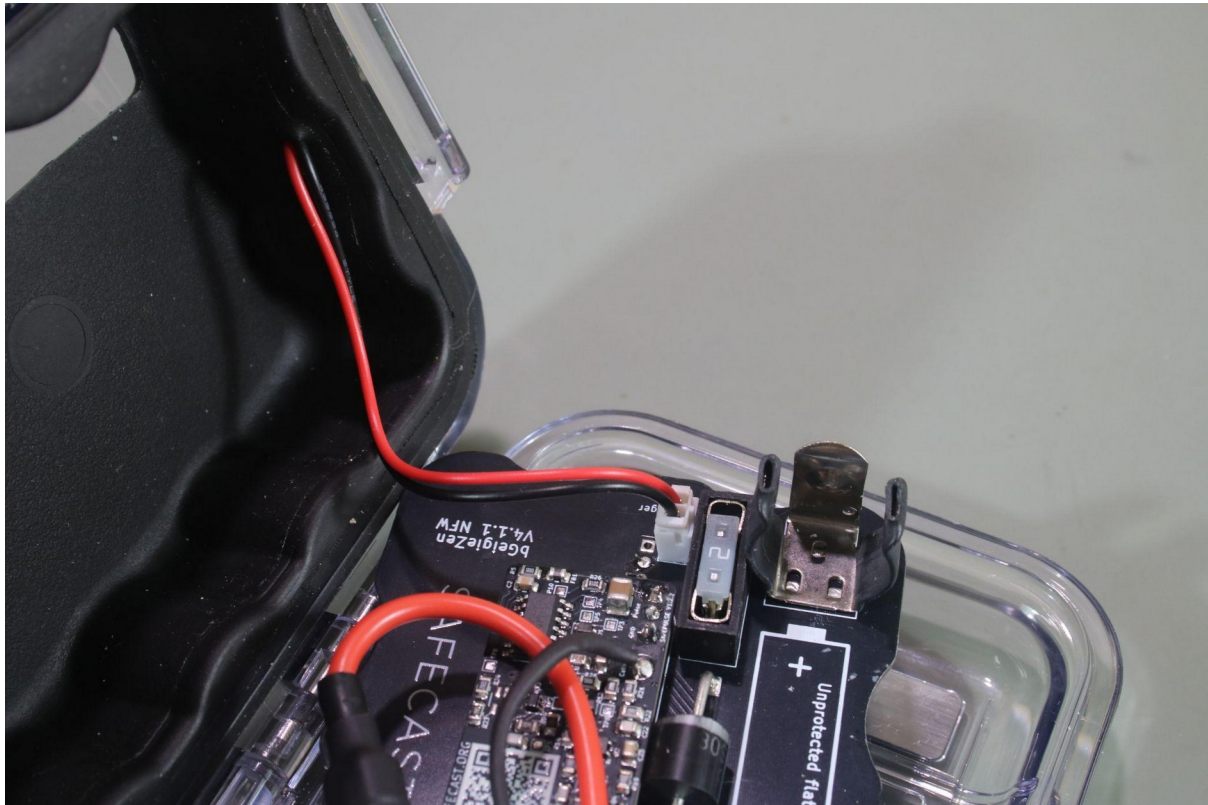


Figure 29: The battery wire from the charger module connects through the rubber liner and is plugged into the circuit board. Observe the correct polarity.

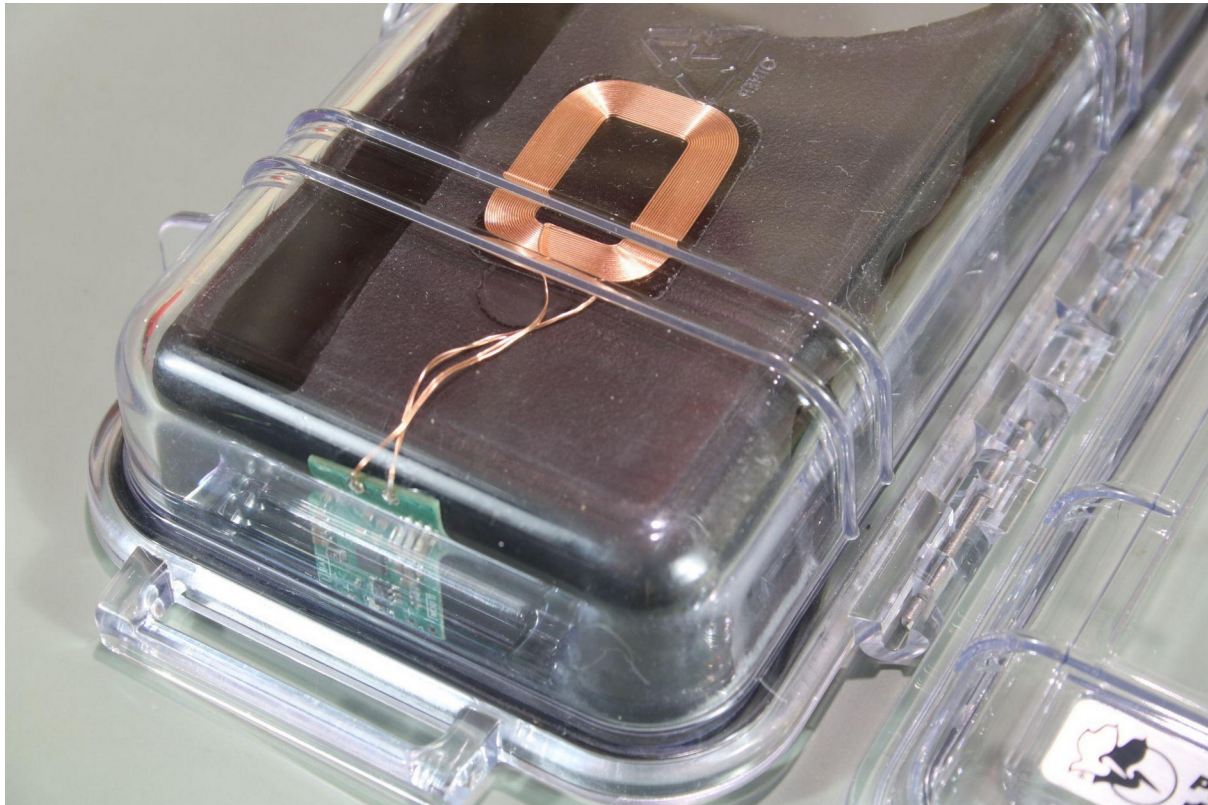


Figure 30: Qi charger module and coil held in place by contact with the case bottom.

Complete assembly

Insert the preprogrammed MicroSD card into the slot at the top of the controller module. Insert the battery into the battery holder (**observe the + – polarity**). Place the completed assembly into the Pelican 1015 Micro Case and latch the lid shut. The fit is quite tight, ensuring that the device does not rattle in the case (Figure 31, Figure 32).



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Figure 31: Complete assembly



Figure 32: Bottom view

Powering up and testing:

Power on by opening the case and holding the power button (right side of controller) for a half second. The bGeigieZen should display its ID number on the boot-up screen. Hold the power button for three seconds to shut down.

Happy Safecasting!

Refer to the operating instructions [[WHERE?]]

Quick start manual <https://t.ly/PKFL5>

User manual at <https://t.ly/9vQqN>